

QuickBite - Software Architecture & Design Document

Milestone 2 Submission

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Project: QuickBite Food Delivery System

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1. High-Level Architecture

1.1 Use Case Diagram

Use Cases

High Priority:

- Browse Restaurants
- Search Restaurants/Menu
- Login
- Register Account
- Track Delivery
- Confirm Order
- Modify Order
- Review Payment
- Create Order
- View Menu
- Rate Order

Medium Priority:

- Manage Menu
- Register Restaurant
- Update Account
- Receive Notifications
- Complete Delivery
- Cancel Order
- Mark Out for Delivery

Low Priority (Optional):

- View All Orders
 - Write Review
 - Request Refund
 - View Reviews
 - Generate Reports
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1.2 Architecture Overview

We chose MVC because it separates the user interface (View) from the business logic (Model). This allows our team (Julian, Vinay, Branson, Leo) to work on the front end and back end in parallel without conflicts, accelerating development. This is also known as the Model, View, Controller architecture, and is how our system will interact with itself.

Model (Data & Business Logic) – Responsible for managing the data, state, and DB interaction. It also enforces rules of the application.

View (User Interface) – Responsible for displaying information from the Model to the user and sending user actions to the controller.

Controller (Intermediary) – Acts as the intermediate middle man for the view and model, receiving user input from the view, and sending to the model for interactions with the DB, and then selects the view for rendering.

With our specific system, an example would include the login feature, where users are displayed a login button (from the view), clicking/interacting with the button (controller), the controller sends the login information to the model where the data is validated (model), and then sent back to the controller, where the correct view is displayed based on the state of the controller (success login/fail login).

With a non-functional requirement (e.g., supporting 500 concurrent users), MVC supports NFR-2 (Scalability) by decoupling the View from the Model. This allows us to optimize the database queries (Model) independently of the user interface, ensuring that heavy back end processing does not freeze the user's browser experience.

2. Low-Level Design

2.1 Component Identification

Feature 1: Main Class is `UserCreation`, its responsibilities are to handle account creation.

- Supporting two roles: Customer (the one ordering food) and Restaurant owner
- Storing user information after it's created
- Validating username and password and updating it
- Login system

Feature 2: Main Class is `RestaurantMenu`, its responsibility is ensuring correct information about the restaurant are stored inside the system.

- Storing restaurant information including menu and location of the restaurant
- Restaurant owners can edit their menu and display its availability

Feature 3: Main Class is `MenuBrowsing`, its responsibility is filter out restaurant given customer's requested food options.

- Query system for customer to choose their order
- Matching customer's searched item to database
- Limiting the restaurant pages to display
- Navigation system for page requested by the customer

Feature 4: Main Class is `OrderManager`, its responsibility is to handle customer's order.

- Order creation by the customer and correctly storing order information
- Following business rules based on location
- Modification of order by customer and restaurant owner
- Updates on order status

- Canceling orders within certain time

Feature 6: Main Class is PriceCalculator, its responsibility is calculate total order cost.

- Properly handle tax and currency based on location
- Tips system
- Display the right currency (location)

Feature 7: Main Class is PaymentHandler, its responsibility is validate customer payment.

- Validating information of customer's payment method
- Accepting the payment method and rejecting them

Feature 8: Main Class is NotificationManager, its responsibility is notify the customer or restaurant about the order.

- Notification of the order placed to customer and notify the restaurant about the order
- Notifying the customer of status of their order (Pickup by the courier, delivered, delays, cancellations)
- Notifying issues about the customer's order from the restaurant

Feature 9: Main Class is UserReview, its responsibility is display review of certain restaurants.

- Rating system is 1star-5star review
- Only those that have completed their order can rate
- Calculate the average review from the customers

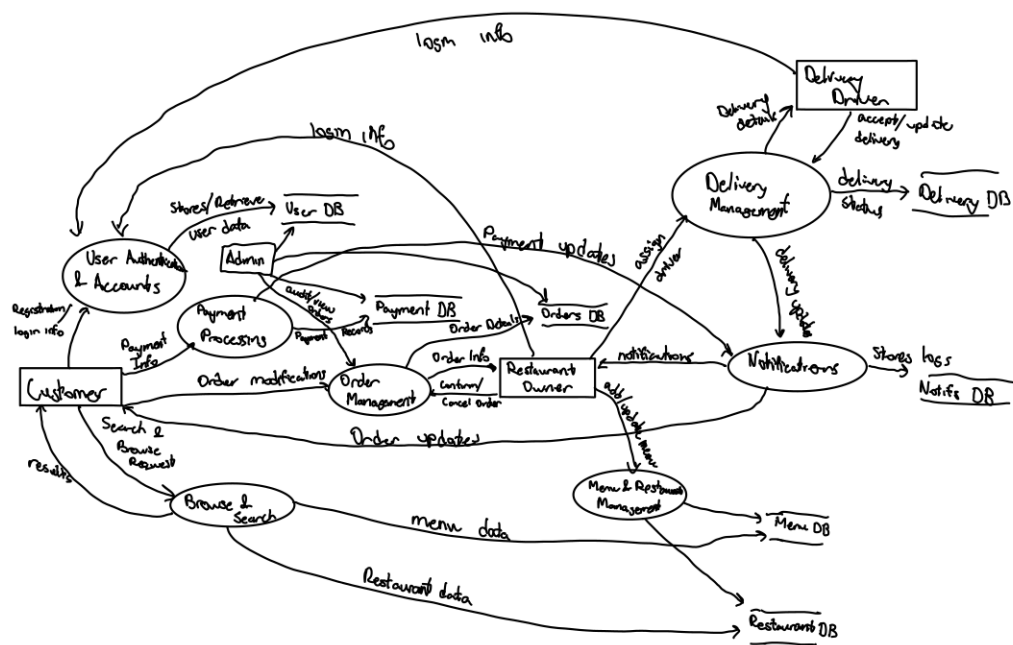
Connections

- Feature 2 gives information for Feature 4 to handle the right order for the customer
 - Feature 6 handles the amount deduction so Feature 7 can deduce the right amount for customer's valid information
 - Feature 7 can validate customer's payment to make Feature 4 deny or accept the customer's order
 - Feature 3 should accurately grab information from Feature 2
 - Feature 9 allows store review to corresponding restaurant (Feature 2)
 - Feature 8 acts as messenger to customer about their order (Feature 4)
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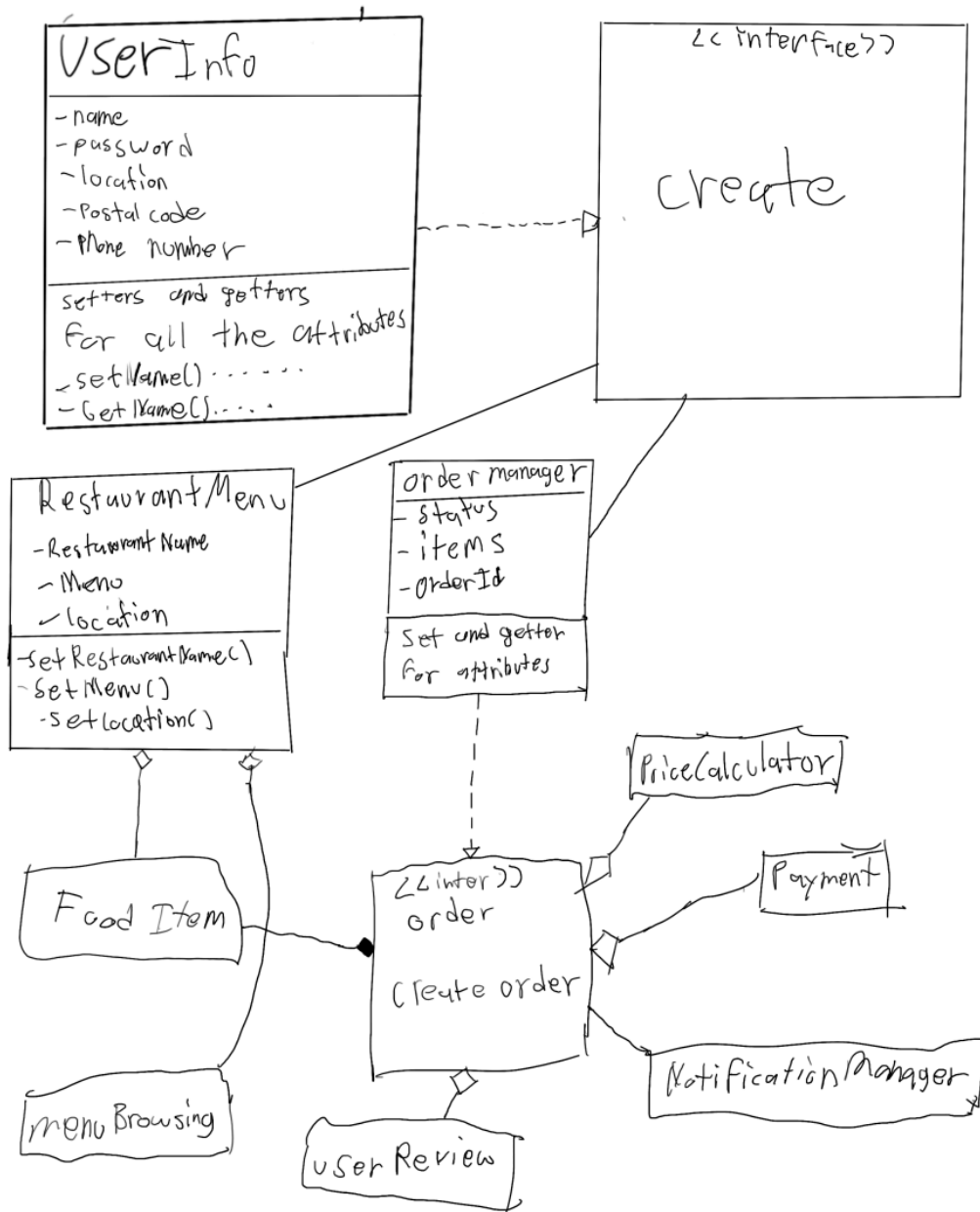
2.2 Design Diagrams

2.2.1 Data Flow Diagram (Level 1)

QuickBite Data Flow Diagram: Level 1

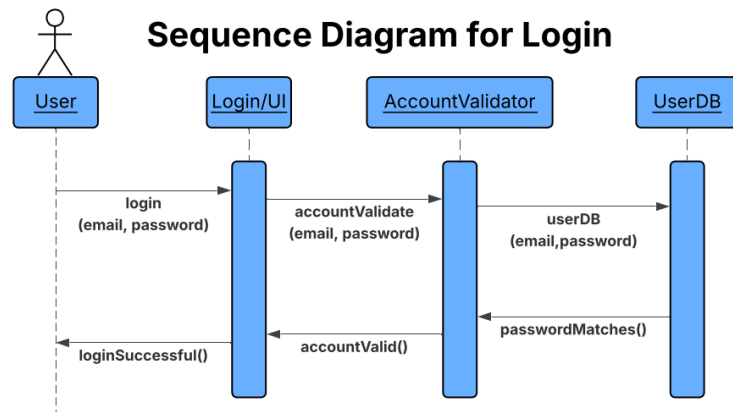


2.2.2 Class Diagram



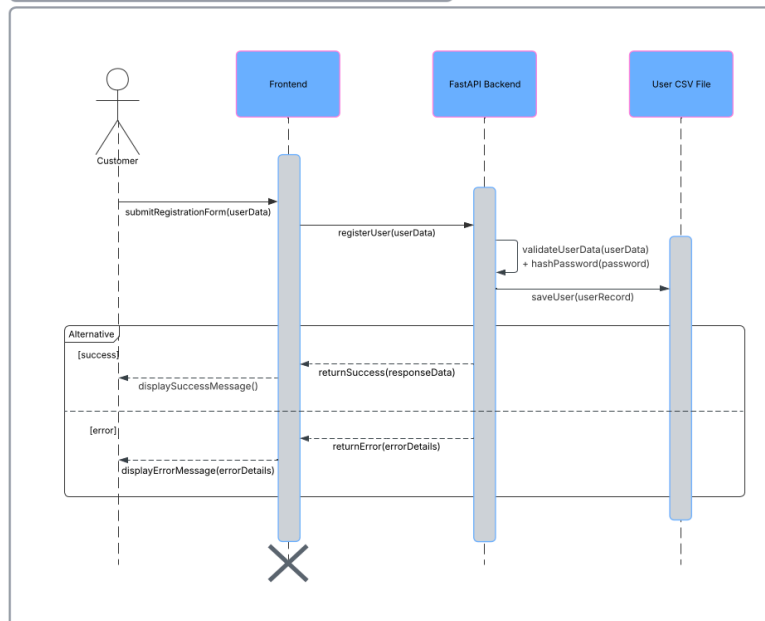
2.2.3 Sequence Diagrams

Sequence Diagram 1: User Login

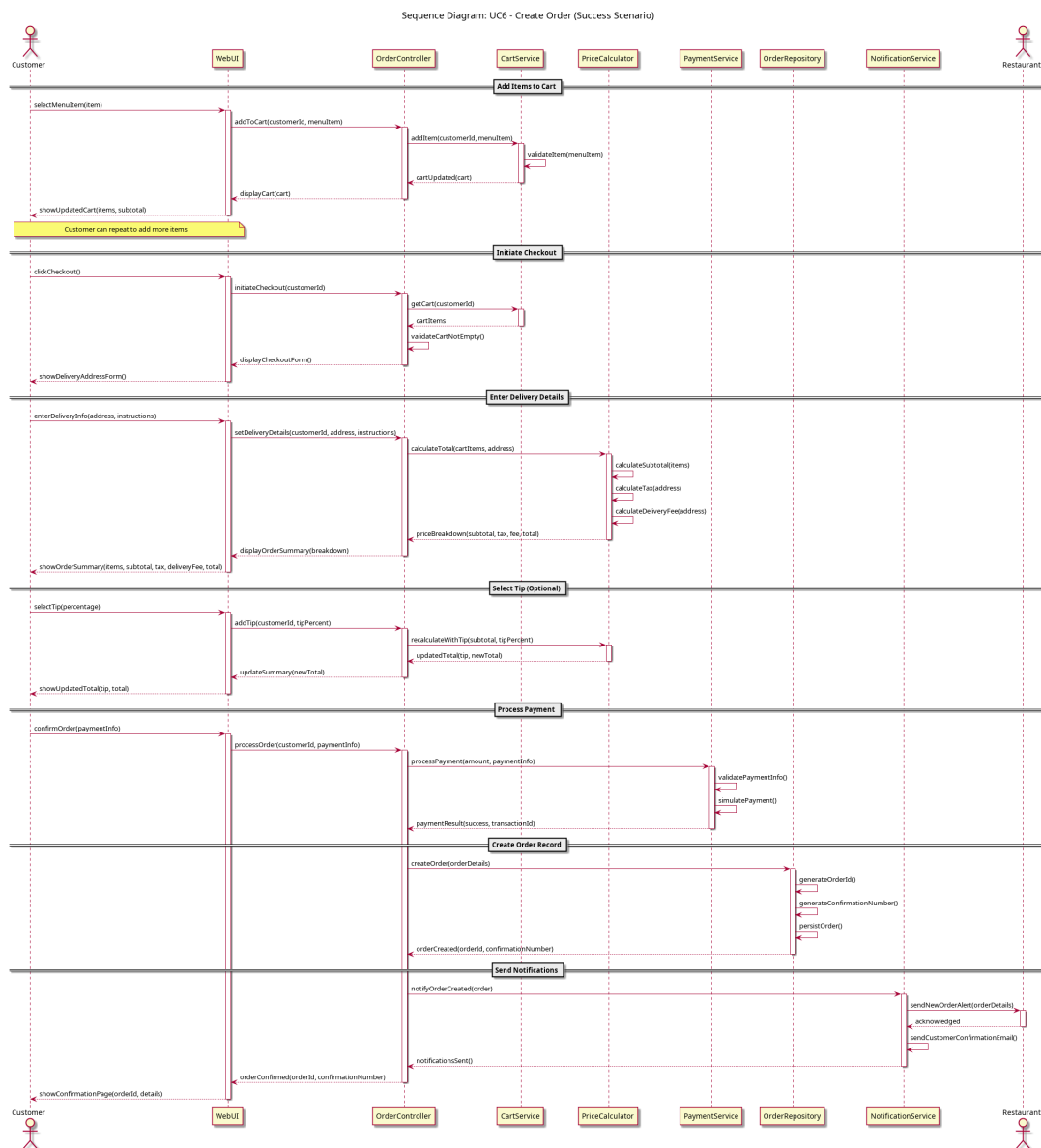


Sequence Diagram 2: User Registration

Registration sequence diagram



Sequence Diagram 3: Create Order (Place Order)



3. Traceability Matrix

The following matrix extends Milestone 1 by adding system components to the corresponding requirements and user stories.

High Level Req	Description	System Req	Requirement	User Stories	Components
HR1	User Registration & Authentication	SR1	The system shall support two users roles: Customer and Restaurant Owner/Manager	Feat1-US1, Feat1-US3	UserCreation

		SR2	The system will store user information after their account has successfully been created	Feat1-US1	UserCreation
		SR3	The system will support user authentication by using username and password	Feat1-US2	UserCreation
		SR4	The system will support account updates to user information	Feat1-US1	UserCreation
HR2	Menu Storing	SR5	The system shall store restaurant information	Feat2-US1	RestaurantMenu
		SR6	The system shall store menu information associated with a specific restaurant	Feat2-US1, Feat2-US2	RestaurantMenu
		SR7	The system shall ensure that all menu items are correctly connected to an existing restaurant	Feat2-US2, Feat2-US3	RestaurantMenu
		SR8	The system shall enforce data validation rules for all restaurant and menu data	Feat2-US3	RestaurantMenu
		SR9	The system shall prevent the entry of invalid or missing values for required data fields	Feat2-US2, Feat2-US3	RestaurantMenu
		SR10	The system shall allow restaurant owners to manage	Feat2-US1,	RestaurantMenu

			their restaurant and menu information	Feat2-US2	
HR3	Menu Browsing	SR11	The system will perform queries based on customer search terms	Feat3-US1	MenuBrowsing
		SR12	The system will return a temporary list of restaurants that matches search terms	Feat3-US1	MenuBrowsing
		SR13	The system will implement navigation to a restaurant's page	Feat3-US1	MenuBrowsing
HR4	Order Management	SR14	The system shall allow customers to create a new order by adding items from a restaurant's menu	Feat4-US1	OrderManager
		SR15	The system shall accurately store all necessary order details	Feat4-US1, Feat4-US3	OrderManager
		SR16	The system shall allow customers to modify an order as long as order is pending or in cart	Feat4-US1	OrderManager
		SR17	The system shall lock an order from further customer modification once its status changes to "confirmed"	Feat4-US2	OrderManager
		SR18	The system shall maintain and update the order status through the delivery process	Feat4-US2, Feat4-US3	OrderManager

		SR19	Allow both the customer and restaurant to cancel an order	Feat4-US1, Feat4-US2	OrderManager
HR5	Delivery Management	SR20	The system shall store delivery information with associated orders	Feat5-US1	OrderManager
		SR21	The system shall allow assignment of deliveries to orders	Feat5-US2	OrderManager
		SR22	The system shall track delivery status (assigned, in-transit, delivered)	Feat5-US1	OrderManager
		SR23	The system shall store delivery address and delivery instructions	Feat5-US1	OrderManager
		SR24	The system shall update delivery status as part of the order flow	Feat5-US2, Feat5-US3	OrderManager
		SR25	The system shall record delivery completion time	Feat5-US3	OrderManager
HR6	Price Calculation	SR26	The system shall calculate the tax of the order based on location of customer	Feat6-US1	PriceCalculator
		SR27	The system shall calculate delivery fee based on distance travelled	Feat6-US2	PriceCalculator
		SR28	The item price should display accurately based on location	Feat6-US2	PriceCalculator
		SR29	The system will calculate tips based on the customer's	Feat6-US3	PriceCalculator

			total order before tax		
HR7	Payment Processing	SR30	The system shall simulate payment processing	Feat7-US1	PaymentHandler
		SR31	The system shall follow the correct workflow for accepting or rejecting a payment	Feat7-US1, Feat7-US2	PaymentHandler
		SR32	The system shall update order status based on payment success or failure	Feat7-US1, Feat7-US3	PaymentHandler
		SR33	The system shall generate a confirmation number upon successful payment	Feat7-US3, Feat7-US4	PaymentHandler
		SR34	The system shall handle simulated payment failures and display appropriate error messages	Feat7-US2	PaymentHandler
		SR35	The system shall record payment status with each order	Feat7-US3	PaymentHandler
HR8	Notification Manager	SR36	Exclusive offers will be presented to the user	Feat8-US1	NotificationManager
		SR37	The customer will notified as soon they order	Feat8-US2	NotificationManager
		SR38	The customer is notified if the order is picked up by courier	Feat8-US2	NotificationManager
		SR39	The customer is notified if the	Feat8-US2	NotificationManager

			restaurant has taken the order		
HR9	Reviews & Ratings	SR40	The system shall allow rating of completed/delivered orders on a scale of 1-5	Feat9-US1	UserReview
		SR41	The system shall allow customers to write text reviews for restaurants	Feat9-US3	UserReview
		SR42	The system shall only allow reviews of completed orders they placed	Feat9-US1, Feat9-US3	UserReview
		SR43	The system shall calculate a weighted average of the restaurants reviews	Feat9-US2	UserReview
		SR44	The system shall display reviews on the restaurants page sorted by most recent	Feat9-US2	UserReview

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